

Distributional Modeling and Lagged Dynamics for EEG-Based Behavioral and Trait Prediction

Ayana Mussabayeva^{1*}

ayana.mussabayeva@mbzuai.ac.ae

Anuar Aimoldin^{1*}

anuar.aimoldin@mbzuai.ac.ae

Yedige Mussabayev²

ymussabayev@hof-university.de

Xue Liu^{1, 3}

steve.liu@mbzuai.ac.ae

Kun Zhang^{1, 4}

kun.zhang@mbzuai.ac.ae

¹Mohamed bin Zayed University of Artificial Intelligence (MBZUAI), Abu Dhabi, UAE

²Hof University, Hof, Germany

³McGill University, Montreal, QC, Canada

⁴Carnegie Mellon University, Pittsburgh, PA, USA

*Equal contribution

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Abstract

Robust EEG decoding remains challenging due to low signal-to-noise ratio and pronounced distribution shift across subjects, sessions, and acquisition setups. We study robust EEG decoding under cross-subject and cross-release distribution shift on Healthy

Brain Network EEG recordings from the contrast change detection paradigm. Our central idea is that making task-relevant temporal structure explicit improves robustness under EEG heterogeneity, and we apply it to two complementary benchmark tracks: behavioral decoding (trial-wise reaction time, RT) from stimulus-locked 2 s windows and trait prediction (a subject-level externalizing psychopathology factor) from randomly sampled 2 s segments, where supervision is even weaker.

For behavioral decoding, we replace scalar regression with temporal segmentation: models predict an event-time distribution using Gaussian-smoothed targets and distribution-matching losses, followed by a soft-argmax readout and out-of-fold stacking over distributional meta-features. This framing is broadly useful for brain-behavior decoding when targets are inherently temporal. It aligns supervision with latency variability, enables latent-consistent jittered cropping, and provides an uncertainty signal for more robust ensembling under cross-subject or session shift. We also report competitive lightweight regression baselines under the same subject-disjoint protocol as reference points for ablations. In our experiments, improvements were more sensitive to objective design and distribution-aware ensembling than to moderate increases in model capacity within the tested regime.

For trait prediction, we make temporal coupling explicit with an interpretable lagged dynamics descriptor pipeline based on multi-lag cross-correlations and ridge-stabilized transition operators. This offers a general approach for weakly supervised subject-level decoding from short segments by summarizing dynamical coupling patterns in a compact, interpretable way that can be more stable across recording heterogeneity.

Across both benchmark tracks, competitive performance is achieved with lightweight models and low-dimensional descriptors, supporting the view that objective design and inductive bias grounded in explicit temporal structure can dominate parameter count under heterogeneous EEG and cross-release shift.

1 Introduction

Electroencephalography (EEG) is one of the most widely used techniques for measuring human brain activity, largely because it is noninvasive, comparatively inexpensive, and offers millisecond-scale temporal resolution. These properties make EEG a natural sensing modality for closed-loop and interactive systems, including brain–computer interfaces (BCIs), where decisions must be made real-time based on ongoing neural dynamics. EEG-driven BCIs have been investigated for decades, from early communication systems (Farwell and Donchin, 1988) to applications in neurorehabilitation (Hamedi et al., 2016), real-time neuromodulation (Zrenner et al., 2018), and interactive entertainment (Lécuyer et al., 2008).

Motivated by the success of foundation models in language and vision, the EEG community has recently seen a rapid growth of large-scale pretrained backbones that aim to learn transferable, task-agnostic representations from heterogeneous EEG corpora. Examples include transformer-based pretraining such as EEGPT (Wang et al., 2024), large multi-dataset tokenization and masking strategies such as LaBraM (Jiang et al.,

2024) and EEGFormer (Chen et al., 2024), state-space model backbones such as EEG-Mamba (Wang et al., 2025), and large-scale cross-setup pretraining explicitly targeting montage and recording variability such as REVE (Ouahidi et al., 2025). Autoregressive pretraining has also been explored for generalist EEG modeling (Yue et al., 2025). Overall, these efforts suggest that large-scale self-supervised pretraining can reduce reliance on task-specific labels and improve performance.

Despite this progress, robust EEG decoding remains challenging in practice due to low signal-to-noise ratio (SNR) and strong distribution shifts. EEG recordings reflect mixtures of neural sources through volume conduction and are easily contaminated by non-neural artifacts such as eye movements, facial and neck muscle activities, cardiac signals, and environmental noise. EEG is also highly nonstationary across time and context: changes in alertness, movement, task engagement, and electrode impedance can substantially shift the data distribution even within a single session. Crucially for machine learning, there is substantial inter-subject variability driven by anatomy, development, cognitive strategy, and recording conditions. Together, these factors often yield models that perform well within-subject or within-session but degrade sharply when evaluated across subjects, sites, or dataset releases.

In this work, we argue that a practical route to robustness under such heterogeneity is to make temporal structure explicit, rather than treating each example as a static input-output regression problem. We study two complementary settings where supervision is weak and SNR is low, and where temporal structure is naturally informative. First, we consider behavioral decoding: trial-wise reaction time prediction, where behavior is expressed through event timing within a stimulus-locked window. Second, we consider trait prediction: a stable, participant-level externalizing psychopathology factor from short segments, where the label signal is even weaker and must be captured through consistent temporal coupling patterns. Given the low-SNR and cross-subject/cross-release shift, we deliberately prioritize lightweight models and low-dimensional descriptors, focusing on inductive bias (explicit temporal structure) and robust combination strategies rather than scaling model size.

We evaluate these ideas using the EEG Foundation Challenge benchmark (Aristimounha et al., 2025), which stress-tests generalization under cross-subject and cross-release shift. The benchmark is built on the Healthy Brain Network EEG dataset spanning thousands of pediatric and adolescent participants and multiple experimental paradigms (Shirazi et al., 2024). We focus on the Contrast Change Detection (CCD) paradigm and its two associated prediction problems: trial-wise reaction time and subject-level externalizing.

For reaction-time decoding, we reframe scalar regression as temporal segmentation: instead of predicting a single reaction-time value directly, we supervise models to produce an event-time distribution over the response window using Gaussian-smoothed targets and distribution-matching losses. This formulation supports temporally consistent augmentation, yields uncertainty signals useful for calibration and ensembling, and improves robustness under jitter and distribution shift. For externalizing prediction, we propose a complementary and interpretable lagged dynamics descriptor pipeline based on

multi-lag correlations and ridge-stabilized transition operators, making temporal coupling explicit in a low-dimensional representation. Together, these methods operationalize a shared principle: explicitly modeling temporal structure, either event timing for behavior or lagged coupling for traits, can stabilize learning under low SNR and heterogeneous EEGs.

Finally, in this benchmark setting we observe that strong held-out performance can be achieved with lightweight models and lightweight descriptors. While this is not a controlled scaling analysis, it is consistent with the view that under heterogeneous EEG and cross-release shift, objective design and distribution-aware ensembling can be more influential than moderate increases in model capacity within the tested regime.

2 Related Work

2.1 Robust EEG Decoding under Shift

A central obstacle for EEG decoding is that the data distribution changes substantially across subjects, sessions, and recording setups. Long before the current wave of large pretrained EEG backbones, the BCI community developed methods explicitly aimed at mitigating this shift through feature spaces and normalization procedures designed to be more stable than raw sensor-space signals. A prominent line of work represents trials by spatial covariance matrices and performs classification using the geometry of symmetric positive definite (SPD) matrices, yielding strong cross-subject and cross-session performance in multiple paradigms (Barachant et al., 2012). Building on this, transfer learning work in EEG-BCIs has emphasized distribution alignment and the reuse of data across subjects/sessions to reduce or avoid subject-specific calibration (Jayaram et al., 2016; Zanini et al., 2018; He and Wu, 2020). These approaches underscore a recurring theme: robustness often benefits from inductive biases (e.g., geometry-aware representations, explicit alignment) that reduce nuisance variability before or during learning.

More recently, deep-learning approaches to robustness have imported ideas from domain adaptation, including learning domain-invariant representations by adversarial objectives (Ganin et al., 2016) or by matching second-order statistics across domains (Sun and Saenko, 2016). In EEG, these ideas are typically instantiated as subject- or session-level alignment modules, discrepancy losses, or adversarial heads that encourage invariance to subject identity while retaining task-relevant information. In parallel, self-supervised and large-scale pretraining has emerged as another route to transferability: contrastive or masked objectives can learn reusable representations across tasks and hardware (Kostas et al., 2021), and this direction has expanded rapidly into multi-dataset EEG foundation models (Wang et al., 2024; Jiang et al., 2024; Chen et al., 2024; Wang et al., 2025; Ouahidi et al., 2025; Yue et al., 2025).

Our work is complementary to both alignment-centric transfer learning and large-scale pretraining. Rather than proposing a new invariant backbone or an explicit domain-matching loss, we focus on a task-driven principle: *make task-relevant temporal structure*

explicit. The NeurIPS EEG Foundation Challenge defines two complementary tasks ((EEG, 2025; Aristimunha et al., 2025)). For Challenge 1, this means explicitly modeling *when* the behavioral event occurs within the stimulus-locked window; for Challenge 2, it means explicitly summarizing *lagged coupling* patterns that are plausibly more stable across heterogeneous recordings than instantaneous sensor values. In this sense, we aim to obtain robustness from objective/representation design that exposes temporally meaningful structure, while keeping the core models lightweight.

2.2 Behavioral Decoding as Event Timing / Time-to-Event

Reaction time (RT) and related behavioral measures are inherently temporal: even when evaluation uses a single scalar per trial, the latent object of interest is an *event time* within a post-stimulus window. Cognitive neuroscience has a long tradition of modeling response times as distributions governed by latent decision processes (e.g., accumulation-to-bound models), and EEG analyses often distinguish stimulus-locked from response-locked components, where response-related dynamics can shift in latency across trials. Early work on single-trial ERP analysis and visualization emphasized this point by sorting trials by RT and revealing response-locked structure that is blurred by averaging (Jung et al., 1998). In parallel, supervised RT prediction from single-trial EEG has been explored using classical features or deep models, typically treating RT as direct regression or coarse discretized classification (Chowdhury et al., 2020).

From a machine-learning perspective, reframing scalar regression as distributional prediction has several precedents. Label distribution learning (LDL) treats supervision as a distribution over labels rather than a single point value, providing a natural language for “soft” targets and distribution-matching losses (Geng, 2016). Time-to-event (survival) modeling similarly targets the distribution of event times, classically via hazard-based models (Cox, 1972) and more recently via deep models that directly parameterize discrete-time event distributions (Lee et al., 2018). Although our RT setting is not survival analysis in the clinical sense (there is no censoring mechanism in the benchmark protocol), the conceptual alignment is useful: predicting an event-time distribution makes timing explicit, supports calibration/uncertainty analysis, and can be more compatible with temporal jitter and cropping than point regression.

Our segmentation formulation adopts this perspective in a minimal way tailored to short EEG windows: we supervise a probability mass function over discretized post-stimulus time bins using smoothed targets and distributional losses, then extract RT via a differentiable readout. This bridges single-trial EEG traditions that highlight response-locked latency variability (Jung et al., 1998) with modern distributional supervision frameworks (Geng, 2016; Lee et al., 2018), while remaining computationally lightweight.

2.3 Trait Prediction from Temporal Coupling / Connectivity Descriptors

Predicting stable traits or psychopathology dimensions from short, weakly labeled EEG segments is challenging because the label signal is not tied to a specific within-window event. One influential line of work links externalizing-related psychopathology to reliable EEG/ERP markers, especially reduced P300 amplitude and related time–frequency signatures. A substantial literature supports P3 amplitude reduction as an endophenotype indexing risk for substance use and related externalizing outcomes (Iacono et al., 2003; Iacono and Malone, 2011; Patrick et al., 2006; Hicks et al., 2007). These findings suggest that trait-relevant information can be expressed through stable dynamical properties (e.g., how neural populations coordinate over time), not only through transient event-locked responses.

A natural way to operationalize such dynamical information is through measures of functional and directed connectivity. Functional connectivity metrics (coherence, phase synchronization, phase-slope index, etc.) are widely used but come with well-known interpretational pitfalls in EEG, including volume conduction, common reference effects, and sample-size bias; tutorial reviews emphasize careful choice of metrics and controls (Bastos and Schoffelen, 2015). To mitigate spurious “zero-lag” coupling due to mixing, measures such as imaginary coherency and phase-lag index (PLI) down-weight instantaneous correlations that can arise from common sources (Nolte et al., 2004; Stam et al., 2007). For directed dependencies, autoregressive modeling and Granger-style predictability provide a statistical notion of directed influence (Granger, 1969; Barnett and Seth, 2014; Seth et al., 2015), though their assumptions and sensitivity to preprocessing must be respected in neurophysiological settings.

Our externalizing pipeline sits in this connectivity tradition but is deliberately constrained for robustness and interpretability under short windows and cross-release shift. Instead of estimating high-variance spectral connectivity or full multivariate causal graphs, we summarize lagged temporal coupling using (i) multi-lag cross-correlations and (ii) ridge-stabilized transition operators that capture predictable lag structure in a low-dimensional form. The goal is not to claim mechanistic connectivity, but to extract coupling descriptors that are stable enough to generalize across heterogeneous recordings and informative for subject-level traits that have known EEG correlates (Iacono et al., 2003; Iacono and Malone, 2011).

3 Dataset and Benchmark Tracks

We use the NeurIPS EEG Foundation Challenge benchmark (EEG, 2025), based on curated releases of the HBN-EEG dataset (Shirazi et al., 2024), to evaluate robustness under cross-subject and cross-release distribution shift. The benchmark provides a labeled *development* release and a fully hidden *held-out* release, for which inference and scoring are performed server-side on the challenge platform. We consider two benchmark tracks: (i) trial-wise reaction time prediction (Challenge 1; behavioral

decoding) and (ii) subject-level psychopathology factor prediction (Challenge 2; trait prediction). The releases comprise recordings from more than 3,000 participants aged 5–21 collected across a standardized battery of six EEG tasks spanning both passive and active paradigms (Shirazi et al., 2024; EEG, 2025).

HBN-EEG is distributed in a Brain Imaging Data Structure (BIDS)-compatible format with detailed event annotations using Hierarchical Event Descriptors (HED), enabling reproducible preprocessing and time-locked analyses (Shirazi et al., 2024). In the original acquisition, EEG was recorded in a sound-shielded environment using a high-density 128-channel EGI Geodesic HydroCel system at 500 Hz (online 0.1–100 Hz bandpass), referenced at Cz, with electrode impedances maintained below 40 k Ω . For the challenge, the organizers provide a harmonized derivative of these recordings that is band-pass filtered (0.5–50 Hz) and downsampled to 100 Hz to standardize modeling across releases (EEG, 2025). In the released derivatives, a total of 129 channels are included (128 EEG channels plus a reference); throughout this work we train models on the 128 EEG channels and exclude the reference channel from the learnable input.

In addition to EEG and demographics, the releases include continuous psychopathology dimensions derived from parent-reported Child Behavior Checklist (CBCL) data via a confirmatory bifactor model: a general p -factor alongside orthogonal specific factors for internalizing, externalizing, and attention-hyperactivity (HBN, 2024). In this work, we focus on the externalizing factor, which captures variance related to behavioral dysregulation (e.g., rule-breaking/aggressive and conduct-like problems) (HBN, 2024).

One of the active paradigms in HBN-EEG and the focus of both benchmark tracks is the contrast change detection (CCD) task. In each CCD trial, participants viewed two concentric, flickering grating stimuli (one left-tilted and one right-tilted). After a variable delay, the contrast of one stimulus increased, making it perceptually dominant. Participants were instructed to respond as quickly as possible to indicate whether the dominant grating was left or right-tilted, after which they received immediate accuracy feedback (smiley for correct, sad face for incorrect). The CCD recordings support two complementary prediction settings that emphasize different temporal structure: event-locked response timing for trial-wise behavior and segment-level temporal coupling for subject-level traits.

3.1 Challenge 1: Reaction Time Prediction in CCD

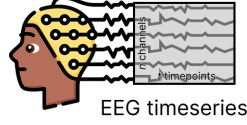
The goal of Challenge 1 is behavioral decoding via prediction of the reaction time (RT) of the participant’s button press on each CCD trial. The model is provided with a fixed-length 2 s EEG segment (128 channels at 100 Hz) time-locked to the stimulus event. In the competition’s official inference setup (including the held-out release), the 2 s input window corresponds to the interval from 0.5 s to 2.5 s after stimulus onset (Figure 1); trials whose responses fall outside this interval are excluded by the benchmark construction. The target is a single scalar RT per trial, and performance is evaluated using normalized RMSE (nRMSE), defined as RMSE divided by the standard deviation of the true targets (EEG, 2025). Because the label reflects when a response occurs within the post-stimulus interval, this prediction problem naturally motivates modeling RT as

Challenge 1:

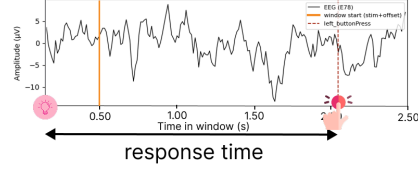
Input: 2 seconds of 100Hz EEG (128 channels)

Prediction: stimulus response time

Contrastive change
detection task
visual stimulus



EEG timeseries



Challenge 2:

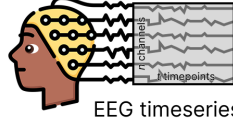
Input: 2 seconds of 100Hz EEG (128 channels)

Prediction: psychopathology factors

(scalar values, showing state of mental health)



p-factor
externalizing factor
internalizing factor



EEG timeseries

Figure 1: EEG Foundation Challenge tracks. Both tracks use 2 s EEG window (100 Hz, 128-channel) from CCD. Challenge 1 predicts trial-wise response time from a stimulus-locked 0.5–2.5 s window, while Challenge 2 predicts subject-level psychopathology factors (externalizing) from 2 s CCD segments that are not event-locked.

an event-time quantity rather than a static scalar regression target.

3.2 Challenge 2: Externalizing Factor Prediction

Challenge 2 targets trait prediction through the participant-level psychopathology dimension given by the externalizing factor score derived from questionnaire measures using a bifactor model (privacy-preserving, orthogonal factor scores) (EEG, 2025). The input is a 2 s EEG segment from the CCD recordings of a participant (i.e., not necessarily stimulus-locked to a specific event). The benchmark construction yields a set of 2 s segments per participant; during training we treat these segments as a pool and sample segments stochastically to reduce redundancy while preserving the same supervision signal (all segments from a participant share the same target). The model must map these short windows to the participant’s externalizing score. Since many segments share the same subject label, supervision at the 2 s segment level is weak, which motivates descriptors that capture consistent coupling patterns rather than event-locked responses. The primary difficulty is out-of-distribution robustness: models must generalize across unseen participants and across releases with potential shifts in recording conditions and subject composition.

4 Behavioral Decoding: Reaction Time Prediction

4.1 Evaluation Protocol and Data Partitions

Within the labeled development release, we create a fixed subject-disjoint split into a training partition $\mathcal{D}_{\text{train}}$ and a held-out validation partition $\mathcal{D}_{\text{val}}$ (80/20 by subjects) and keep it fixed across all experiments (see Table 1). Base models are trained on $\mathcal{D}_{\text{train}}$ only. $\mathcal{D}_{\text{val}}$ is used for checkpoint selection / early stopping, selecting per-model temperature τ^* (Eq. 5, Sec. 4.4), and training the stacking meta-learner via subject-disjoint cross-validation (Section 4.6). Final reported benchmark scores are computed on the fully hidden held-out release $\mathcal{D}_{\text{holdout}}$ (evaluated server-side).

Partition	Role (who sees what)
$\mathcal{D}_{\text{train}}$ (dev, 80% subjects)	Train base models.
$\mathcal{D}_{\text{val}}$ (dev, 20% subjects)	Early stopping / checkpoint selection; select per-model τ^* ; train stacker via 5-fold subject-disjoint CV (Out-of-Fold (OOF) evaluation on $\mathcal{D}_{\text{val}}$).
$\mathcal{D}_{\text{holdout}}$ (held-out release, labels hidden)	Organizer scoring / reporting only; no training or tuning.

Table 1: Evaluation protocol and data partitions.

4.2 Baseline: Lightweight Temporal ConvNet for Direct RT Regression

We begin with a lightweight baseline that treats Challenge 1 as direct reaction-time regression: a stimulus-locked 2 s EEG crop $X \in \mathbb{R}^{C \times T}$ with $T = 200$ samples (100 Hz) is mapped to a scalar RT.

Consistent with the low-SNR and distribution shift setting, we focus on lightweight backbones that are less prone to overfitting and enable extensive ablations, calibration, and ensembling.

Several lightweight models (see `models/regression`) were trained using a two-stage schedule (base training followed by fine-tuning), where fine-tuning restarts from the best validation checkpoint with a reduced learning rate on plateaus, and with strong augmentations (time cutout, channel dropout, additive Gaussian noise, time-scaling, and mixup).

Key Challenge 1 training and inference settings are summarized in Table 7, Sec. 6. These models serve as reference points for ablations and for isolating the effect of making temporal structure explicit in Section 4.4. In contrast to the segmentation formulation, the direct-regression setup tended to require more careful tuning and was less stable under heavy temporal jitter and cropping. This baseline treats the window as a static input-output regression problem, without an explicit representation of response timing.

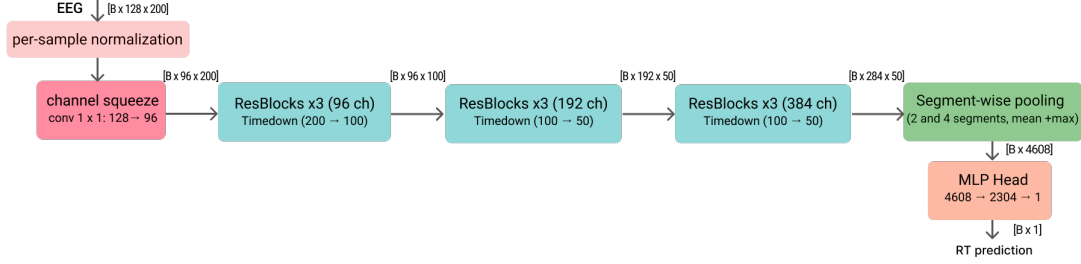


Figure 2: SneddyNet: direct RT regression baseline architecture. A lightweight 1-D temporal ConvNet maps a stimulus-locked 2 s EEG crop (128 channels, 100 Hz) to a scalar reaction-time prediction. Per-sample normalization and a learned 1×1 channel-squeeze layer ($128 \rightarrow 96$) are followed by residual depthwise-separable stages with anti-aliased temporal downsampling, segment-wise pooling (2 and 4 segments; mean+max), and an MLP head.

The main regression backbone (Figure 2) SneddyNet is a lightweight 1-D temporal ConvNet with: (i) per-sample normalization to reduce cross-subject amplitude variation, (ii) a learned channel-squeeze layer (1×1 conv, $128 \rightarrow 96$) that forms data-driven electrode mixtures, and (iii) an EfficientNet-like temporal hierarchy with widening stages of residual depthwise-separable blocks ($96 \rightarrow 192 \rightarrow 384$ channels) and anti-aliased temporal downsampling. Coarse timing cues are retained using segment-wise statistical pooling: the time axis is split into 2 and 4 segments, mean and max are computed per segment, and the resulting features are passed to a small MLP head.

The initial RT regression baseline (SneddyNet) achieved **0.9360 nRMSE** on the benchmark’s fully hidden held-out release (evaluated server-side; Table 2, Sec. 4.3), representing a substantial improvement over generic off-the-shelf architectures implemented in the Braindecode library (Schirrneister et al., 2017). In our RT regression setup, the best-performing Braindecode baselines (e.g., EEGNet (Lawhern et al., 2018) and EEGConformer (Song et al., 2023)) tended toward near-constant predictions and did not improve over the mean-predictor reference, yielding $\text{nRMSE} \approx 1.00$.

Despite strong performance and fast training, direct regression does not explicitly represent the event-timing structure of the task. Under heavier temporal jitter and cropping, optimization tended to over-focus on local waveform micro-structure rather than learning *when* the response-related event occurs. This limitation motivated the segmentation reformulation in Section 4.4, which makes response timing explicit via an event-time distribution.

4.3 Direct Regression Baselines: Model Zoo and Takeaways

Although our final RT solution uses temporal segmentation, we found it important to document strong regression-style baselines (including scalar and time-distribution readouts) as a reference point and ablation. Table 2 summarizes representative regression-style baselines trained and evaluated under the same validation protocol.

All baselines implement the same high-level pipeline: (a) per-sample temporal

Table 2: Direct Reaction Time baselines (validation). Lower nRMSE is better. *DS* = depthwise-separable.

Model	Key design (high-level)	Temporal readout	nRMSE ↓	Size (MB)
SneddyNet-S	Compact DS-residual encoder Shallow depth and small kernels emphasize local temporal motifs. Strong temporal compression yields coarse features suited to pooling.	StatPool + MLP	0.9299	4.95
SneddyNet-L	Higher-capacity DS-residual encoder Wider/deeper stack and larger kernels increase temporal context and channel capacity. Same coarse temporal hierarchy, but with stronger representational power.	StatPool + MLP	0.9152	45.66
SneddyRTNet-Dilated	Dilated DS-residual backbone Dilations expand receptive field while delaying temporal compression relative to SneddyNet. Designed to support distributional timing rather than coarse pooling.	TimeDist soft-argmax	0.9181	49.24
RecurTiny EE-GRT (ACT)	Tied-weight recursive DS core Iterative refinement with step conditioning; adaptive computation via ACT halting. Large effective depth at low parameter cost, paired with distributional timing.	Recur-ACT TimeDist	0.9179	1.67

standardization, (b) a learnable 1×1 channel-mixing projection ($C \rightarrow c_0$), and (c) a lightweight depthwise-separable residual convolutional backbone inspired by DS-ResNet style and MobileNet/Xception (Howard et al., 2017; Chollet, 2017). They differ mostly in how the final temporal feature map is converted into an RT estimate, that is, the *temporal readout*. In all cases, supervision remains scalar RT (or an RT-derived target) without explicit event-time labels; TimeDist is used only as a readout parameterization, not as event-time supervision.

We consider three temporal readout families. **StatPool+MLP** (SneddyNet family) applies segmented mean and max pooling over time to encode coarse early and late cues (conceptually similar to segment-based aggregation in TSN (Wang et al., 2016)), followed by an MLP regressor. **TimeDist soft-argmax** (SneddyRTNet) predicts a distribution over time bins and reads RT as its expectation via a soft-argmax (coordinate-regression style) readout (Nibali et al., 2018). **Recur-ACT TimeDist** (RecurTinyEEGRT) refines the time distribution iteratively using tied weights with adaptive computation time (ACT) style halting (Graves, 2016) and FiLM conditioning (Perez et al., 2018), producing a halting-weighted mixture distribution before the same expectation readout. SneddyRTNet additionally uses dilated convolutions to expand receptive field while delaying temporal compression relative to SneddyNet (Yu and Koltun, 2016).

We observe two patterns emerge. First, increasing capacity improves StatPool+MLP performance (SneddyNet-L vs. SneddyNet-S). Second, time-distribution readouts can match large regressors while being substantially more parameter-efficient (RecurTinyEEGRT at 1.67 MB). However, despite being competitive, these direct regression baselines

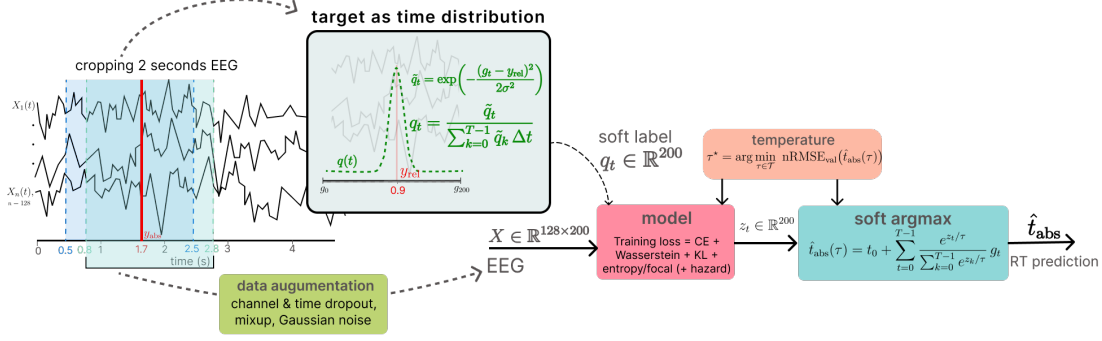


Figure 3: Temporal segmentation pipeline for RT prediction. For each trial we cache a longer stimulus-locked segment (4 s in our implementation) and sample a 2 s subwindow with start-time jitter for training; the RT label is shifted into the sampled window and a Gaussian soft target q_t is formed. Per-time logits z_t are produced by a 1-D segmentation model and are optimized with distribution-matching losses (e.g., soft-label CE, CDF-based Wasserstein-1, KL, and mild entropy penalty); focal reweighting and a hazard head were explored but not used in the final pipeline. RT is obtained by temperature-scaled soft-argmax, with τ^* selected to minimize validation nRMSE.

are less robust under heavy temporal jitter and cropping. Moreover, calibration under cross-subject and cross-release shift is not guaranteed by the readout alone; obtaining well-calibrated uncertainty typically requires additional modeling and/or post-hoc calibration (Guo et al., 2017; Kendall and Gal, 2017). These considerations motivate the segmentation formulation below, which makes response timing explicit via event-time supervision and distribution-matching losses.

4.4 Reformulating RT Prediction as Temporal Segmentation

To make event timing explicit and to enable strong crop augmentation, RT prediction was reformulated as *temporal segmentation*: instead of regressing a scalar directly, we predict a distribution over time bins and read out RT as its expectation (Figure 3). Compared to distributional readouts used only as a final head, the key difference here is that supervision is defined in time coordinates and shifted consistently under crop jitter, so augmentation preserves the event-time target by construction. In practice, this produced a large performance gain and faster, more stable convergence, while making crop augmentation safer because the supervision explicitly encodes *where* the event occurs. This is our RT instantiation of explicit temporal structure: behavior is modeled through event timing rather than static regression.

Notation and crop sampling. Let a 2 s crop be $X \in \mathbb{R}^{C \times T}$ with sfreq = 100 Hz, $T = 200$ and $\Delta t = 1/\text{sfreq}$. For training-time augmentation, we retain a longer stimulus-locked segment of length 4 s (0–4 s post-stimulus) and sample a 2 s subwindow from it, introducing temporal jitter while preserving trial identity. The crop start is clipped

so that the labeled response remains inside the sampled 2 s window (label-preserving jitter). During validation and official inference, we always use the fixed stimulus-locked 2 s window with offset $t_0 = 0.5$ s (0.5–2.5 s), so that $t_{\text{abs}} = t_0 + t_{\text{rel}}$.

Soft target distribution. For a sampled crop, the ground-truth event time within the crop is represented as a relative time $y_{\text{rel}} \in [0, \text{crop_sec})$ with $\text{crop_sec} = 2$. A smooth segmentation label is constructed using a Gaussian kernel with bandwidth σ :

$$\tilde{q}_t = \exp\left(-\frac{(g_t - y_{\text{rel}})^2}{2\sigma^2}\right), \quad q_t = \frac{\tilde{q}_t}{\sum_{k=0}^{T-1} \tilde{q}_k}, \quad \sum_{t=0}^{T-1} q_t = 1. \quad (1)$$

This smoothing stabilizes optimization while preserving temporal localization; in practice, σ can be annealed (start larger and decrease over training).

Model output and losses. A 1-D segmentation backbone (Section 4.5) produces logits z_t for $t = 0, \dots, T-1$, and a predicted event-time distribution is obtained via temperature-scaled softmax,

$$p_t(\tau) = \frac{e^{z_t/\tau}}{\sum_{k=0}^{T-1} e^{z_k/\tau}}. \quad (2)$$

Training minimizes a weighted segmentation objective:

$$\begin{aligned} \mathcal{L} = & \lambda_{\text{time}} \text{RMSE}(\hat{t}_{\text{rel}}(\tau), y_{\text{rel}}) + \lambda_{\text{CE}} \text{CE}(q, p(\tau)) \\ & + \lambda_{W1} W_1(\text{CDF}(q), \text{CDF}(p(\tau))) + \lambda_{\text{KL}} D_{\text{KL}}(q \| p(\tau)) - \lambda_H H(p(\tau)), \end{aligned} \quad (3)$$

where

$$\text{CE}(q, p) = -\sum_{t=0}^{T-1} q_t \log p_t, \quad D_{\text{KL}}(q \| p) = \sum_{t=0}^{T-1} q_t \log \frac{q_t}{p_t}, \quad H(p) = -\sum_{t=0}^{T-1} p_t \log p_t.$$

The time-RMSE term is the primary objective, since minimizing the readout error aligns directly with the benchmark metric. The remaining terms act as optional distribution-level regularizers on $p(t)$: (i) $\text{CE}(q, p)$ encourages mass near the soft target; (ii) the CDF-based Wasserstein-1 term penalizes temporal misalignment in a geometry-aware way (shifting probability mass along time rather than matching bins independently); (iii) $D_{\text{KL}}(q \| p)$ discourages off-target modes by penalizing probability mass far from q ; and (iv) the entropy penalty $-H(p)$ encourages sharper (more confident) predictions when distributions become overly diffuse under strong augmentation. We also explored focal reweighting and a hazard-parameterized head (mapping logits to a discrete-time hazard and then to an event-time distribution), but these variants did not yield consistent validation improvements under the subject-disjoint protocol and are not used in the final pipeline. We treat augmentation strength jointly with temperature calibration because it affects predictive confidence.

Inference and temperature tuning. The relative event time is estimated by expectation (soft-argmax),

$$\hat{t}_{\text{rel}}(\tau) = \sum_{t=0}^{T-1} p_t(\tau) g_t, \quad \hat{t}_{\text{abs}}(\tau) = t_0 + \hat{t}_{\text{rel}}(\tau), \quad (4)$$

where $g_t = t \Delta t$ denotes the discrete time grid. Under squared error, the Bayes-optimal point estimate is the conditional mean; accordingly, reading out RT by $\mathbb{E}_{p(t)}[g]$ aligns naturally with the nRMSE objective (RMSE up to a constant normalization).

In practice, different training choices (loss weighting, augmentation strength, and checkpoint selection) produced time-distributions with markedly different sharpness and entropy: some models generated very peaked $p(t)$ while others were systematically more diffuse. This creates a confidence-scale mismatch, where naive averaging (or stacking without temperature-scaled inputs) can underweight accurate but underconfident models and overweight overconfident ones.

We therefore apply temperature scaling (Guo et al., 2017) to tune the softness of $p(t)$, improving single-model readout and making confidence profiles more comparable across models, which in turn stabilizes ensembling and stacking. Temperature is selected by validation tuning for each base model m :

$$\tau_m^* = \arg \min_{\tau \in \mathcal{T}_{\text{cal}}} \text{nRMSE}_{\text{val}}(\hat{t}_{\text{abs}}^{(m)}(\tau)). \quad (5)$$

In our implementation, $\mathcal{T}_{\text{cal}}$ is a 30-point linear grid in $[0.4, 1.8]$ (Table 7, Sec. 6).

Because confidence scales differ across architectures and augmentation settings, this step is critical; mixup in particular tends to increase entropy and shift the optimal τ .

4.5 Segmentation Architecture: SneddyUNet Family

A family of efficient 1-D encoder-decoder networks (“SneddyUNet”) is used to map a stimulus-locked EEG crop $X \in \mathbb{R}^{C \times T}$ to per-time logits $z_{0:T-1}$ (Figure 4). All segmentation variants follow a 1-D U-Net style encoder-decoder with skip connections for precise localization (Ronneberger et al., 2015). The encoder progressively downsamples the temporal axis to build multi-scale context, while the decoder upsamples and fuses encoder features to recover per-time outputs at the original 200-sample resolution (internal down/up-sampling is used only to form multi-scale features). Each scale is refined with lightweight residual blocks (ResNet-style skips) (He et al., 2016) built from depthwise-separable 1-D convolutions for computational efficiency (Howard et al., 2017; Chollet, 2017). To mitigate aliasing under temporal decimation, downsampling applies a fixed depthwise smoothing filter followed by strided averaging (“blur-pool” style) (Zhang, 2019). A dilated bottleneck further enlarges the receptive field without additional downsampling (Yu and Koltun, 2016).

Segmentation model zoo. Table 3 lists the variants used for ensembling/stacking. SneddyUNet-L/M/S share the same SneddyUNet1D template and vary width and the number of residual refinement blocks per level. AttnSneddyUNet replaces the bot-

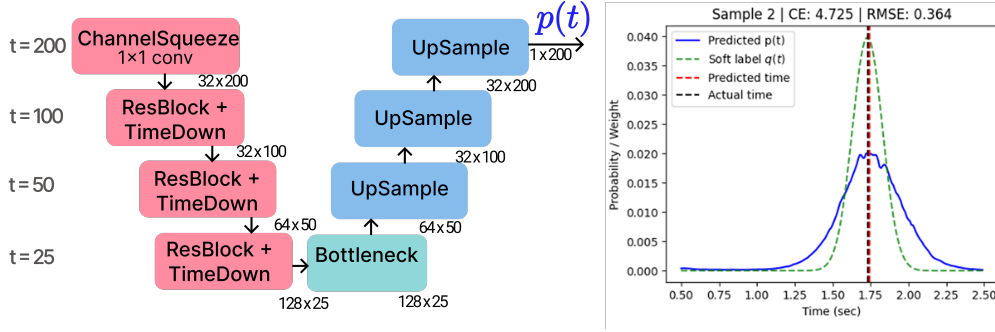


Figure 4: SneddyUNet1D for RT segmentation. A lightweight 1-D U-Net maps a 2 s, 128-channel EEG crop to per-time logits and an event-time distribution $p(t)$. The encoder applies a 1×1 channel-squeeze layer followed by residual blocks with temporal downsampling (TimeDown); a bottleneck aggregates context, and the decoder upsamples with skip connections to recover time-resolved outputs. The right panel shows an example prediction $p(t)$ and Gaussian soft target $q(t)$, together with the resulting predicted vs. true response time.

tleneck with temporal multi-head self-attention (MHSA) (Vaswani et al., 2017) and uses stochastic depth (DropPath) regularization (Huang et al., 2016) to stabilize training at higher capacity. EEGInceptionSeg replaces single-path temporal filtering with Inception-style multi-branch temporal kernels (Szegedy et al., 2015), following the EEG-Inception design for ERP feature extraction (Santamaría-Vázquez et al., 2020). Finally, SneddyUNet-FM augments the channel-mixing front-end with factorized cross-channel interactions inspired by Factorization Machines (FM) (Rendle, 2010), optionally also injected within encoder stages to introduce a distinct cross-channel inductive bias.

Architectural diversity is introduced to improve ensembling and stacking. SneddyUNet-L (wide) uses a wider encoder and 5 residual blocks per level, providing the highest capacity. SneddyUNet-M retains the same depth (5 blocks/level) with a more moderate width for a stronger accuracy–efficiency trade-off, while SneddyUNet-S uses a lighter encoder (3 blocks/level) and yields faster, lower-variance base learners. Long-range temporal interactions are emphasized in AttnSneddyUNet (MHSA) via a multi-head self-attention bottleneck. EEGInceptionSeg replaces single-path temporal filtering with multi-branch kernels for multi-scale feature extraction. Finally, SneddyUNet-FM augments the standard channel-mixing front-end with an FM-style factorized interaction module (optionally also injected within encoder stages), providing a distinct cross-channel inductive bias.

4.6 Out-of-Fold Stacking over Time Distributions

Segmentation-style outputs enable richer ensembling than scalar averaging, but combination becomes sensitive to model confidence: naive averaging can overweight overconfident models and underweight accurate but underconfident ones. We therefore train a second-stage stacker on distributional meta-features extracted from the time-distribution outputs.

Table 3: Segmentation model variants used for reaction-time prediction. Each model outputs per-time logits and is trained under the same segmentation objective; architectural diversity is exploited for ensembling/stacking.

Model	Architecture	Key features	Strengths
SneddyUNet-L (wide)	SneddyUNet1D	wider encoder; depthwise-separable convs; 5 Res-Blocks/level; dilated bottleneck	highest capacity; captures complex local temporal patterns
SneddyUNet-M	SneddyUNet1D	moderate width; depthwise-separable convs; 5 Res-Blocks/level; dilated bottleneck	best single-model accuracy / efficiency tradeoff
SneddyUNet-S	SneddyUNet1D	lighter encoder; depthwise-separable convs; 3 Res-Blocks/level; same bottleneck family	fast, low-variance; complements deeper models
AttnSneddyUNet (MHSA)	Attention U-Net	MHSA bottleneck (4 heads, depth=3); DropPath; positional depthwise conv	captures long-range temporal dependencies
EEGINceptionSeg	EEG-Inception 1D	multi-branch temporal scales (0.5/0.25/0.125 s) with feature fusion	strong multi-scale feature extraction
SneddyUNet-FM	Factorized U-Net	FM-style cross-channel interactions at input; optional per-stage FM injection	distinct cross-channel inductive bias

All base segmentation models are trained once on the training partition $\mathcal{D}_{\text{train}}$ (Table 1, Sec. 4). We then compute their logits on the held-out validation partition $\mathcal{D}_{\text{val}}$ and convert logits to meta-features (e.g., peak time, entropy/variance, temperature-conditioned expected times, and CDF quantiles) to train a second-stage stacker. For server-side inference on $\mathcal{D}_{\text{holdout}}$, we compute the same meta-features from held-out logits and apply the fold-trained stackers (median-aggregated; Figure 5).

Meta-features from logits and temperature-scaled probabilities. Meta-features are extracted per model from (i) raw logits and (ii) temperature-scaled probabilities computed for multiple temperatures $\tau \in \mathcal{T}_{\text{meta}}$. From logits, a hard-peak estimate and confidence proxies are computed:

$$t^* = \arg \max_t z_t, \quad \hat{t}_{\text{hard}} = t_0 + g_{t^*}, \quad z_{\text{max}} = \max_t z_t, \quad m_{\text{top2}} = z_{\text{max}} - \max_{t \neq t^*} z_t. \quad (6)$$

For each $\tau \in \mathcal{T}_{\text{meta}}$, probabilities are formed as $p(\tau) = \text{softmax}(\mathbf{z}/\tau)$ (Eq. 2, Sec. 4.4), where $\mathbf{z} = (z_0, \dots, z_{T-1})$ denotes the per-time logit vector for a given model and window. Distributional summaries are computed, including the soft-argmax time (Eq. 4, Sec. 4.4),

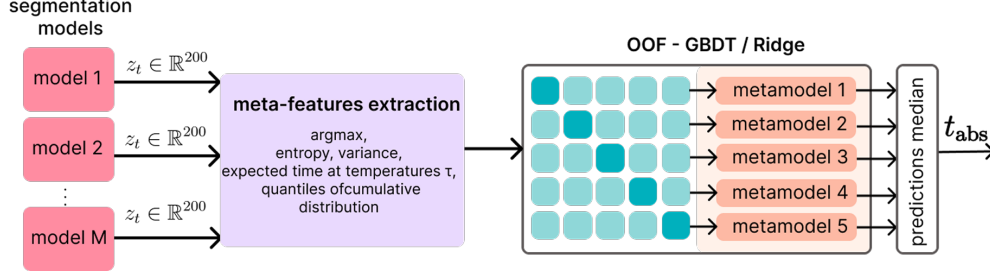


Figure 5: OOF stacking for RT prediction from time-distribution outputs. Per-model logits/distributions are converted into meta-features (e.g., peak time, entropy/variance, temperature-conditioned expected times, and CDF quantiles) and combined by a second-stage stacker (Ridge/GBDT). Predictions from fold-trained stackers are aggregated at inference (median) to produce the final RT estimate.

entropy $H(p(\tau))$, peak probability $p_{\max}(\tau)$, time variance $\text{Var}_{p(\tau)}[g]$, and CDF time-quantiles (10/50/90%). In our implementation, $\mathcal{T}_{\text{meta}} = \{0.7, 1.0, 1.3\}$ (Table 7, Sec. 6).

Meta-learner and inference. We use linear Ridge regression as a strong baseline meta-learner, and gradient boosting to capture non-linear reliability patterns across models and temperatures. The meta-learner is trained only on $\mathcal{D}_{\text{val}}$ using 5-fold subject-disjoint cross-validation: each subject is assigned to exactly one fold (Figure 5). For fold k , the stacker is fit on the other four folds and predicts RT for the held-out fold. We report validation nRMSE using these out-of-fold predictions. Folds are stratified to match the per-subject mean RT distribution across folds. For inference on $\mathcal{D}_{\text{holdout}}$, the same feature construction is applied to held-out logits; predictions from the fold-trained stackers are aggregated (median) to obtain the final RT estimate. Manual blending is retained as a reference, but it cannot adapt weights to sample-specific confidence profiles.

4.7 RT Prediction Results: Base Models and Stacking

Segmentation models are first evaluated individually with and without temperature tuning. Compared to the best direct regression baselines (Table 2, Sec. 4.3, best nRMSE ≈ 0.915), segmentation yields a clear improvement even at the single-model level (Table 4, best temperature-tuned nRMSE ≈ 0.899). This is consistent with the view that making event timing explicit is a stronger inductive bias for RT decoding than scalar regression in this setting. Wider and deeper U-Nets (e.g., SneddyUNet-M) achieve the strongest single-model accuracy while remaining efficient, whereas smaller models (e.g., SneddyUNet-S and EEGInceptionSeg) provide complementary, low-variance predictions that are useful for ensembling. Attention bottlenecks can improve long-range temporal context but come with higher parameter and storage cost.

The performance achieved by each standalone model is presented in Table 4, which also reports the serialized checkpoint size. Notably, the strongest single model (SneddyUNet-M) is only 12.5 MB, and several complementary base learners are below 10 MB (e.g., SneddyUNet-S: 9.2 MB; EEGInceptionSeg: 0.66 MB). This indicates that

Table 4: Representative segmentation models for RT prediction (validation). “def.” uses $\tau = 1$; “cal.” uses the calibrated τ^* selected on the validation split.

Model	nRMSE (def.) ↓	nRMSE (cal.) ↓	τ^*	Size (MB)
SneddyUNet-L (wide)	0.9025	0.9023	1.076	20.8
SneddyUNet-M	0.8989	0.8988	1.076	12.5
SneddyUNet-S	0.9038	0.9038	1.000	9.2
AttnSneddyUNet (MHSA)	0.9015	0.9015	1.028	130
EEGINceptionSeg	0.9174	0.9169	1.124	0.66
SneddyUNet-FM	0.9105	0.9093	1.221	41

Table 5: Ensembling and stacking for RT prediction (validation OOF and hidden held-out release).

Method	Valid nRMSE ↓	Held-out nRMSE ↓
Best temperature-tuned single model	0.898757	0.92334
Manual blending	0.899979	0.91840
Ridge stacking	0.890430	0.91539
Gradient boosting stacking	0.883365	0.91394

competitive RT prediction can be achieved with lightweight architectures, supporting rapid experimentation and resource-constrained deployment.

Ensembling is performed at two levels. First, manual blending averages temperature-tuned single-model predictions and yields performance comparable to the best single model. Second, stacking learns how to combine models based on distributional meta-features (peak location, sharpness, entropy, variance, and CDF quantiles), allowing confidence to be re-weighted in a sample-dependent manner. In this setting, gradient boosting provides the strongest overall performance. Table 5 reports results for different combination strategies on the validation split (stackers evaluated out-of-fold) and on the organizer-held held-out release (labels hidden).

The improvement from stacking is consistent with the intuition that different architectures produce distinct time-distribution shapes (sharp peaks vs. diffuse uncertainty). By conditioning on distributional meta-features, the stacker learns when each model is reliable and adjusts their contributions accordingly, yielding a more stable final RT estimate than fixed-weight averaging.

In the next section, we apply the same overarching principle, explicit temporal structure, to trait prediction, where supervision is weaker and informative signals are captured via lagged temporal coupling rather than event timing.

5 Trait Prediction: Externalizing Factor from Lagged EEG Dynamics

5.1 Evaluation Protocol and Window Sampling

Challenge 2 targets subject-level prediction of the externalizing factor, a continuous psychopathology dimension related to behavioral patterns. Input samples are 2 s EEG windows sampled at 100 Hz (128 channels), drawn from the CCD task recording of each subject. Each window shares the same subject-level target externalizing score y_{ext} , and the goal is to predict $\hat{y}_{\text{ext}}$ from short segments that may occur at arbitrary times within the session. This setting provides our second instantiation of explicit temporal structure: rather than event timing, we model stable traits through lagged temporal coupling patterns. To reduce sensitivity to subject- and site-specific amplitude scaling, per-channel standardization is applied within each window prior to feature extraction (Figure 6).

Although y_{ext} is defined at the subject level, we construct training examples by randomly sampling multiple 2 s windows from each subject’s CCD recording and treating each window as a noisy observation of the same latent trait. To avoid leakage, train/validation splits are performed at the subject level (no subject appears in both splits).

5.2 Lagged Correlation and Transition Representations

Externalizing is hypothesized to reflect differences in dynamical regulation rather than instantaneous amplitude patterns. Accordingly, we seek information in how activity at time t relates to activity at $t + \ell$ across channels. Lagged correlations summarize time-scale dependent co-fluctuations, while ridge transition matrices provide an oriented linear mapping that summarizes asymmetric lagged dependence under a linear approximation. This design is motivated by autoregressive viewpoints in which transition operators encode directed temporal dependencies (e.g., VAR transition matrices) (Geiger et al., 2015). Although no causal claims are made here, these lagged descriptors act as mechanistic proxies for persistence, propagation, and feedback imbalance across the sensor network.

Let $X \in \mathbb{R}^{C \times T}$ denote a per-window normalized EEG segment with $T = 200$ time samples. Let $\Delta t = 1/100$ s and consider lags $\ell \in \mathcal{L}$ measured in samples (e.g., $\ell = \{5, 10, 25, 50, 100\}$ corresponding to 0.05 to 1.0 s). Two complementary matrix representations are computed for each lag.

Lagged cross-correlation matrices. For each lag ℓ , we compute a time-shifted cross-correlation matrix $C^{(\ell)} \in \mathbb{R}^{C \times C}$ with entries

$$C_{ij}^{(\ell)} = \text{corr}(X_{i,1:T-\ell}, X_{j,1+\ell:T}), \quad (7)$$

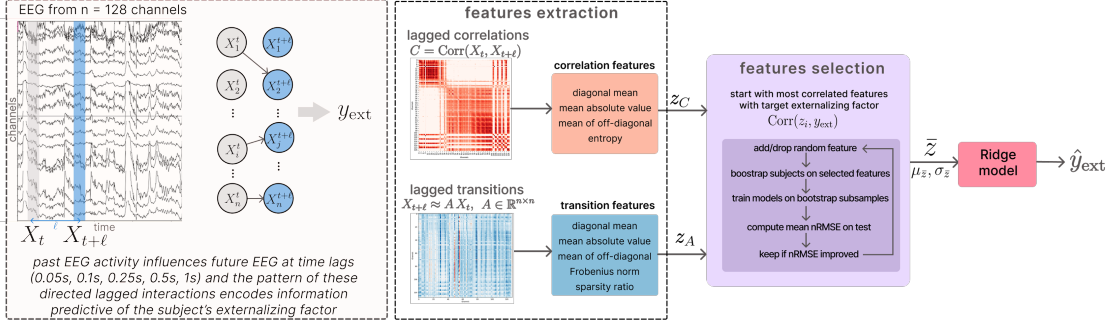


Figure 6: Externalizing prediction pipeline. From a per-window normalized 2 s, 128-channel window, lagged correlations $C^{(\ell)} = \text{corr}(X_t, X_{t+\ell})$ and ridge transition matrices $A^{(\ell)}$ from $X_{t+\ell} \approx A^{(\ell)} X_t$ are computed over multiple lags (0.05 to 1 s). Compact scalar summaries are extracted and a bootstrap add/drop procedure selects stable subsets, after which a ridge regressor outputs $\hat{y}_{\text{ext}}$.

that is, Pearson correlation across time within the window between channel i at time t and channel j at time $t + \ell$. The diagonal captures within-channel temporal persistence, while off-diagonal entries summarize lag-specific cross-channel coupling.

Ridge transition matrices. To summarize lagged dependencies in an oriented linear form, we fit a ridge-stabilized linear transition operator for each lag ℓ :

$$X_{:,1+\ell:T} \approx A^{(\ell)} X_{:,1:T-\ell}. \quad (8)$$

We write the estimator in covariance form. Let $X_t = X_{:,1:T-\ell}$ and $X_{t+\ell} = X_{:,1+\ell:T}$, and define

$$C_{xx} = \frac{1}{T-\ell} X_t X_t^\top, \quad C_{xy} = \frac{1}{T-\ell} X_{t+\ell} X_t^\top. \quad (9)$$

We use a scale-normalized ridge regularizer with

$$\alpha = \text{tr}(C_{xx})/C, \quad (10)$$

and obtain the closed-form solution

$$A^{(\ell)} = C_{xy} (C_{xx} + \lambda \alpha I)^{-1}. \quad (11)$$

This normalization makes λ less sensitive to overall signal power and improves numerical stability when estimating $A^{(\ell)}$ from short windows. The diagonal of $A^{(\ell)}$ reflects self-persistence; off-diagonal entries summarize cross-channel influences; and the antisymmetric component $A^{(\ell)} - (A^{(\ell)})^\top$ provides a compact proxy for directionality and feedback imbalance (without causal interpretation).

5.3 Scalar Feature Construction Across Lags

From each lagged correlation matrix $C^{(\ell)}$ and ridge transition matrix $A^{(\ell)}$, we extract a compact set of scalar summaries to obtain an interpretable, low-dimensional feature vector per lag. This aggregation is motivated by the weak supervision in Challenge 2: many short windows share a single subject-level label, so we emphasize stable lagged coupling descriptors that are less sensitive to window-specific noise.

Summaries from lagged correlations. For the lagged correlation matrix $C^{(\ell)}$, the mean diagonal value captures within-channel temporal stability. Global synchronization is summarized by the mean absolute correlation across all entries, and inter-channel coupling is summarized by the mean off-diagonal correlation (reported both as signed mean and as mean absolute value when used as a magnitude descriptor).

To quantify whether coupling is concentrated or diffuse, we compute the entropy of normalized absolute entries. For a matrix $M \in \mathbb{R}^{C \times C}$, let $v = \text{vec}(|M|)$ and $p = v / \sum_k v_k$ (if $\sum_k v_k > 0$). We define

$$H(M) = - \sum_{k: p_k > 0} p_k \log p_k, \quad (12)$$

so lower entropy indicates a more peaked interaction pattern.

Summaries from ridge transitions. For the lagged transition matrix $A^{(\ell)}$, self-persistence is captured by the mean diagonal value, and overall influence strength by the mean absolute entry magnitude. Cross-channel effects are summarized by the mean off-diagonal value, and the overall magnitude of the lagged mapping by the Frobenius norm $\|A^{(\ell)}\|_F$ as a proxy for mapping strength. To reflect directionality and feedback imbalance, we use the magnitude of the antisymmetric component $A^{(\ell)} - (A^{(\ell)})^\top$ (without causal interpretation).

Finally, we include a sparsity ratio defined as the fraction of entries below a small magnitude threshold:

$$\text{sparsity}_\theta(A^{(\ell)}) = \frac{1}{C^2} \sum_{i=1}^C \sum_{j=1}^C \mathbf{1}(|A_{ij}^{(\ell)}| < \theta), \quad (13)$$

with $\theta = 0.05$ in our implementation.

Across lags, these descriptors summarize how activity persists and propagates across channels as a function of time scale, while remaining comparatively robust to nuisance amplitude variation.

Implementation details. All features are computed on per-window standardized signals (channel-wise z-scoring within each 2 s segment). We use the lag set $\mathcal{L} = \{5, 10, 25, 50, 100\}$ (0.05 to 1.0 s at 100 Hz) and a ridge coefficient λ fixed across lags. To reduce redundancy between highly overlapping windows and control computational cost, windows are subsampled with a fixed stride during feature extraction.

5.4 Bootstrap-Driven Feature Selection and Ridge Regression

Given lagged temporal-coupling descriptors across multiple lags, we train a ridge regression model using a stability-oriented feature selection procedure. Subjects are first partitioned such that the mean and standard deviation of the externalizing target are approximately matched across folds, ensuring comparable target distributions during selection and evaluation. An initial feature pool is created by ranking candidates via absolute Pearson correlation with y_{ext} on the training subjects.

Feature selection is performed with a stochastic add/drop search coupled to bootstrap validation. At each iteration, training subjects are resampled with replacement (bootstrap over subjects rather than windows), a ridge regressor is fit on the current feature subset, and predictions are generated for a fixed subject-level validation split that is not resampled during bootstrapping. Window-level predictions are aggregated to subject-level scores before computing nRMSE, matching the evaluation target.

This yields a distribution of nRMSE values across bootstraps. Candidate feature-set updates are accepted only if they (i) improve median nRMSE and (ii) reduce variability across bootstraps (measured by the standard deviation of nRMSE), favoring subsets that are both accurate and stable. The procedure is repeated until a stopping criterion is met (no accepted update for a fixed number of iterations), after which the final ridge model is refit on the selected feature subset and used for inference.

By representing each 2 s EEG window through lag-dependent correlation structure and regularized linear transitions, the externalizing prediction task is reduced to learning from compact descriptors of temporal stability, coupling strength, asymmetry, and sparsity across multiple physiologically relevant timescales (50 ms to 1 s). Because the descriptors are computed on standardized signals and rely on regularized estimators, sensitivity to site-specific amplitude scaling is reduced, supporting generalization to unseen subjects and heterogeneous dataset releases.

5.5 Externalizing Factor Prediction Results

We first compared against end-to-end EEG baselines trained directly on randomly sampled 2 s CCD segments. We evaluated several off-the-shelf architectures from Braindecode (Schirrneister et al., 2017), including EEGNet (Lawhern et al., 2018) and EEGConformer (Song et al., 2023), trained from scratch with mean-squared error and early stopping based on validation nRMSE. Across architectures and reasonable hyperparameter settings, these end-to-end baselines tended to collapse toward near-constant predictions (close to the training-set mean), yielding performance near the mean-predictor reference (nRMSE ≈ 1). This behavior is consistent with the weak supervision in Challenge 2, where many short windows share the same subject-level label and window-level SNR is low. In contrast, our lagged dynamics descriptor pipeline provides an explicit temporal-structure representation that yields a more stable signal for trait prediction.

Table 6 compares our final lagged EEG dynamics pipeline against (i) a lightweight convolutional regression baseline (SneddyNet) trained directly on 2 s windows and (ii)

Table 6: Externalizing factor prediction results on the official held-out evaluation (lower nRMSE is better; nRMSE = 1 corresponds to predicting the mean target).

Method	Externalizing nRMSE ↓
End-to-end Braindecode baselines (trained from scratch)	≈ 1.00
Regression SneddyNet (direct window regression)	1.0080
NeuroSned (lagged EEG dynamics)	0.98519

representative end-to-end Braindecode baselines trained from scratch.

Overall, explicitly modeling lagged temporal coupling yields the most reliable improvements in this low-SNR, subject-level setting, where end-to-end regression on short windows provides limited effective supervision.

6 Reproducibility and Code Availability

All code (training and inference), notebooks (data preparation, regression/segmentation, calibration, ensembling and stacking), and pretrained weights are available at <https://github.com/sneddy/neurosned>.

Challenge 1 (RT): exact inference reproducibility. We release the exact checkpoints for the base RT models and the fitted stacking meta-models used in our final submission. Running inference with these released artefacts reproduces our submitted RT predictions.

Challenge 1 (RT): training notes. Training uses staged fine-tuning with early stopping on validation nRMSE; on plateaus, we reload the best validation checkpoint and reduce the learning rate (typically by a factor of 0.5). Because this procedure is adaptive and GPU training is not perfectly deterministic, retraining can yield small variations; Table 7 summarizes the default settings used in our released notebooks.

Challenge 2 (externalizing): inference reproducibility. We release the final fitted ridge model used for externalizing prediction together with the feature-extraction code and the selected feature subset. Running inference with these released artefacts reproduces our submitted Challenge 2 predictions.

Challenge 2 (externalizing): training notes. The bootstrap-driven add/drop feature selection is stochastic (bootstrap resampling over subjects); re-running selection can yield small variations in the chosen subset. Our paper specifies the feature construction (lags, per-window standardization, ridge-stabilized transitions) and the selection criterion, and the released notebooks implement the exact pipeline.

Table 7: Minimal reproducibility details for Challenge 1 (RT). These settings summarize the defaults used in our released training notebooks and stacking code.

Component	Setting
Input / preprocessing	2 s window at 100 Hz, 128 EEG channels (reference excluded). Per-sample temporal standardization in-model. Official inference window: 0.5–2.5 s post-stimulus.
Training crop jitter (segmentation)	From a cached stimulus-locked 4 s segment (0–4 s post-stimulus), with probability $p_{\text{crop}}=0.3$ sample a 2 s subwindow with start-time jitter around the default 0.5 s offset (uniform jitter ± 0.25 s), clipped so the labeled event lies within the crop; otherwise use the fixed 0.5–2.5 s crop.
Soft target (segmentation)	Gaussian label smoothing (Eq. 1, 4.4) with $\sigma = 0.15$ s.
Segmentation objective	Primary term: RMSE on soft-argmax time (Eq. 4, Sec. 4.4); auxiliary terms used when beneficial: soft-label CE, CDF-based Wasserstein-1, KL, and mild entropy penalty; focal weighting and a hazard-parameterized head were explored but did not yield consistent gains and were not used in the final pipeline.
EEG augmentations (segmentation)	Time scaling ($p=0.2$, scale 0.8–1.2), channel dropout ($p=0.25$, drop fraction sampled uniformly up to a max value; max 0.8 in our implementation), time cutout ($p=0.25$, length 10–60 samples), Gaussian noise ($p=0.2$, std ≈ 0.01 –0.04), RT-binned mixup ($p=0.1$, Beta($\alpha=0.4$)).
Loss weights (segmentation)	Released defaults: $\lambda_{\text{time}} = 3$, $\lambda_{\text{CE}} = 1$, $\lambda_{W1} = 10^{-3}$, $\lambda_{\text{KL}} = 0$, $\lambda_H = 0$. During development, auxiliary weights were occasionally adjusted (or disabled) when beneficial for validation nRMSE; W_1 uses a smaller coefficient due to its larger raw scale.
Optimization (segmentation)	Adam with initial LR 10^{-3} (typical), early stopping on validation nRMSE; on plateaus, reload best checkpoint and reduce LR (typically by a factor of 0.5; staged fine-tuning). Batch size 2000 in the released notebooks.
Temperature scaling (calibration)	τ^* chosen by grid search over $\mathcal{T}_{\text{cal}}$: 30 values linearly spaced in $[0.4, 1.8]$ (Eq. 5, Sec. 4.4).
Stacking meta-features	Per-model features from logits and from $p(\tau)$ at $\mathcal{T}_{\text{meta}} = \{0.7, 1.0, 1.3\}$: hard peak time, entropy, peak probability/margin, variance, and CDF quantiles (10/50/90%).
Meta-learner	Ridge regression (alpha grid) and histogram gradient boosting (lr 0.05, max_iter 2000, early stopping). 5-fold subject-disjoint CV within the validation partition; fold predictions aggregated by median.

7 Conclusion

We studied robust EEG decoding under cross-subject and cross-release distribution shift using the EEG Foundation Challenge benchmark on HBN-EEG CCD recordings. Our central message is that making temporal structure explicit improves robustness under heterogeneous, low-SNR EEG. We instantiated this principle in two complementary settings: event timing for behavioral decoding and lagged temporal coupling for trait prediction.

For trial-wise RT prediction we reported strong direct regression baselines as reference points and found that objective design, temperature scaling and distribution-aware ensembling provided consistent gains beyond moderate increases in backbone capacity within the tested regime. The obtained results suggest that scalar regression tends to encourage reliance on global window statistics and exploit global texture-like patterns. Reformulating scalar regression as temporal segmentation by supervising an event-time distribution better matched the nature of the problem and enabled time-consistent crop augmentation. Combining several compact 1-D encoder–decoder variants further improved robustness: their diverse uncertainty profiles could be exploited via OOF stacking on simple distributional meta-features. Stacking helps because different architectures produce complementary event-time distributions. Using distributional meta-features, the stacker learns trial-wise reliability and adapts model weights accordingly, outperforming fixed averaging.

For trait prediction lagged dynamics descriptors help under shift because they deliberately compress each window into regularized, low-dimensional summaries of persistence and cross-channel coupling across multiple time scales. Per-window standardization and ridge-stabilized transitions reduce sensitivity to nuisance amplitude differences and short-window estimation noise, yielding a more consistent trait signal than end-to-end window regression in a weak-supervision regime.

Overall, the most reliable improvements arose when the objective and representation exposed what mattered about the signal, event timing for behavior and lagged coupling for traits, rather than relying on end-to-end scalar prediction from short windows. In practice, this favors structure-aligned supervision, calibrated distributional outputs, and combination strategies that exploit complementary uncertainty profiles under distribution shift. At the same time, our conclusions are necessarily conditioned on the specific benchmark design and evaluation protocol, and therefore should be read as evidence about robust prediction under this construction rather than as a definitive statement about underlying neural mechanisms.

In particular, lagged-transition descriptors are best understood as regularized mechanistic summaries of temporal dependence useful for stable decoding, but not equivalent to causal connectivity or directed neural influence. We also can not claim guaranteed transfer to other tasks, montages, preprocessing pipelines, or populations. These considerations motivate extending the present framework with causal representation learning to better characterize the hidden processes that underlie trait prediction and behavioral decoding, and to build adaptive cross-task pipelines that generalize beyond CCD and remain robust across diverse acquisition settings.

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